

RASWANTH MURUGAN

Email: raswanth99@gmail.com | Phone: (+91)7022999197 | Github: [RaswanthMurugan20](#)

OBJECTIVE

Problem Solver with 5+ years of experience across industry and academia as an AI researcher, working on complex problems in biology, navigation, imaging, and autonomous systems. Through this breadth, I've developed not just technical depth in AI, but also the ability to learn to learn, ask the right questions, and find answers to ill-structured problems. I personally enjoy working on hard problems and am passionate about deep tech. I bring a inter-disciplinary mindset suited to any domain that values rigorous thinking and adaptability.

ACADEMIC QUALIFICATION

Indian Institute of Technology, Palakkad (Kerala)

Bachelor of Technology, Computer Science and Engineering.

July 2018 - July 2022

CGPA: 8.66/10.00

WORK EXPERIENCE

AI+Science Lab - Caltech & Argonne Lab

Project: AI-driven D-Handed Enzyme Design

Guide: [Anima Anandkumar](#), [Arvind Ramanathan](#), [Frances Arnold](#)

April '25 - Present

Research Scientist

- **Introduction:** *Frances Arnold pioneered directed evolution—bringing evolution into the lab (in vitro) to design enzymes—which won her the 2018 Nobel Prize. While transformative, the process remains slow and often struggles to design enzymes with the desired function and efficiency. This project takes the next step by moving directed evolution in silico using AI and LLM. As a demonstration, we attempt to redesign the TrpB enzyme to catalyze a non-natural reaction entirely on the computer.*
- Implemented the GRPO fine-tuning pipeline - workhorse for the in-silico directed evolution.
- Developed the SFT pipeline with data from Theozyme, informed sequence designs from RFDiffusion2.
- Assisted implementation of MD simulation workflow to run experiments on catalysis for reward design.

Xie Lab - City University of New York & Weill Cornell Medicine

Project: Inverse Molecular Design to Modulate Single-Cell Transcriptomics

Guide: [Lei Xie](#)

Sep '23 - April'25

Research Scientist, New York

- **Introduction:** *Given a description of a diseased state, can generative models design small molecules that move it toward a healthy one? Inspired by text-to-image models, we built a generative AI pipeline that uses gene-expression profiles of healthy and diseased states as input to synthesize candidate drug-like molecules tailored to the phenotype.*
- Led the development of a generative module translating gene expression to molecules.
- Proposed and implemented conditional latent diffusion models to improve generalization.

Samsung R&D Institute

Project: On-device topology change detection using UWB radar sensor data

Guide: [Vijay Narayan Tiwari](#)

Sep '22 - Sep '23

ML Engineer, Bengaluru

- **Introduction:** *UWB sensors excel at precise localization, enabling innovative solutions like low-light focus enhancement for Galaxy S-series devices. Interestingly, its capabilities can also be leveraged to develop navigation aids for visually impaired users. This project explores the application of ML on UWB radar sensors for depth estimation and topology change detection.*
- Developed a 2D-CNN pipeline for UWB sensor-based depth estimation.
- Employed active learning to iteratively improve model robustness and performance.

Indian Institute of Science

Project: Intelligent Onboard Routing in Stochastic Environments using Transformers

Guide: [Deepak Subramani](#)

Jan'22 - Jan'23

Research Fellow

- **Introduction:** Time-optimal path planning in dynamic ocean velocity fields requires addressing sparse observations and computational constraints. This project explored an end-to-end offline RL approach using transformers to reframe planning as a sequence prediction task, enabling implicit learning of both state estimation and planning for efficient and accurate trajectory optimization.
- Ran experiments validating model efficiency across diverse flow scenarios and targets.
- Discovered appending the target state improved model generalization in multi-target flows.

Indian Institute of Technology, Palakkad

Project: Sparse Reward Propagation using Message Passing

Guide: [Chandrashekar](#)

Dec'20 - Apr'21

Open Ended Lab Project

- **Introduction:** In reinforcement learning, an agent's ability to learn and plan depends, among other things, on how well the reward is defined. As the problem and its environment become more complex, designing an informative reward function becomes difficult. This project uses message passing in graph neural networks to shape these sparse rewards into richer signals, enabling agents to learn more efficiently.
- Implemented Actor-Critic algorithm with online GCN training for sparse reward tasks.
- In a grid-world testbed, The GCN-shaped rewards achieved a 24% reduction in regret compared to the baseline and converged 35% faster to the optimal policy.

Spookfish Innovations

Project: Thermal Face Image Reconstruction into Visible Spectrum

Guide: [Anupriya Balikai](#)

June'20 - July'20

CEO's Intern

- **Introduction:** Face reconstruction from thermal images for face recognition in low light environments poses unique challenges due to sparse data and intricate visual details. This project focused on using neural networks to improve face reconstruction techniques for challenging cases, leveraging insights from computer vision and image processing.
- Designed and implemented two CNN models - one for global face structure reconstruction followed by a local facial features reconstruction.
- Conducted experimental performance analysis on multiple datasets to improve robustness.

PUBLICATIONS

- Amin Tavakoli*, **Raswanth Murugan***, Ozan Gokdemir, Arvind Ramanathan, Frances H Arnold, Anima Anandkumar. **"Self Distillation Fine-Tuning of Protein Language Models Improves Versatility in Protein Design"**. In: Machine Learning in Structural Biology 2025 (Copenhagen, Denmark; San Diego, USA). Paper URL: [Click Here](#).
- Rahul Ohlan, **Raswanth Murugan**, Li Xie, Mohammadsadeq Mottaqi, Shuo Zhang, and Lei Xie. **"MolGene-E: Inverse Molecular Design to Modulate Single Cell Transcriptomics"**. In: International Conference on Machine Learning (ICML) 2024 AI for Science Workshop. 2024. Paper URL: [Click Here](#).
- Rohit Chowdhury, **Raswanth Murugan**, and Deepak Subramani. **"Intelligent Onboard Routing in Stochastic Dynamic Environments using Transformers"**. In: Proceedings of the 2023 International Conference on Autonomous Agents and Multiagent Systems. AAMAS '23. London, United Kingdom: International Foundation for Autonomous Agents and Multiagent Systems, 2023, pp. 1688–1696. ISBN: 9781450394321. Paper URL: [Click Here](#)

SCHOLASTIC ACHIEVEMENTS

- **PhD Admit** (Fully Sponsored) from the City University of New York.
- **Recipient of Summer Research Fellowship-2021** awarded by the Indian Academy of Sciences to nearly 120 nationwide with a 2.5% acceptance rate inclusive of undergraduate and graduate students.
- **Gold medal** among 17 participants at the Inter-IIT tech meet 2018-2019 held at IIT BOMBAY for our Autonomous Pesticide Spraying Robot. organized by **Tata Centre for Technology and Design(TCTD)** .
- **95 percentile** in Jee Advanced among 0.23 million candidates.

- **99.4 percentile** in Jee Mains among 1.3 million candidates.
- **State Rank 10** in National Talent Search Examination by the Government of India, 2015.

OTHER SCHOLASTIC PROJECTS

Project meet.me

Aug'21 - Dec'21

UX team lead, Guide: [Mr. Ramaswamy Krishnan-Chittur\(\(Microsoft\)](#)

IIT Palakkad

- **Introduction:** Developed collaboratively by a team of 27 developers as part of the CS5617 Software Engineering project at IIT Palakkad, meet.me is a collaborative application designed for seamless online meetings, integrating real-time screen sharing and a collaborative whiteboard. [GitHub Repository](#)

TimeSync

Sep'20

Hack MIT

Online

- **Introduction:** Hidden cost of "free" services is our time and attention. TimeSync, developed during HackMIT, connects users with accountability partners to combat screen addiction during the pandemic. Rooted in behavioral insights from "Atomic Habits", it fosters lasting productivity habits through peer collaboration [GitHub Repository](#)

Autonomous Pesticide Spraying Robot

Sep'18 - Dec'18

TCTD Innovation Challenge, [Inter-IIT Tech Meet](#)

IIT Bombay

- **Introduction:** Developed an autonomous pesticide spraying robot addressing challenges faced by small-scale Indian farmers, including labor scarcity, health risks, and inefficiencies in manual spraying. The solution integrates automation, optimized pesticide usage, and adaptable design for various plant sizes and growth stages. For more details please click on this [Link](#)

POSITIONS OF RESPONSIBILITY

• Mathematics Tutoring

Sep'25 - Present

Math Tutoring on Preply

- Tutoring mathematics for grade 1 all to Masters students from China, Denmark, Germany and India.

• Member of People Task Force

Dec'22 - Sep'23

Samsung R&D Institute Bengaluru

- Was among selected 6 in the Advanced Research Group consisting of 150 people to improve the work culture within the group.
- Tasked with planning and organization of cultural events and directly reported and received budget from the vice president of Samsung R&D India Bengaluru.

• Editor of TFS & Interviewer at Alumni Cell

Jan'21 - Dec'21

IIT Palakkad - [The Fleet Street](#), Alumni Cell

- The Fleet Street or TFS is a the student media of IIT Palakkad. As a team of 4 editors, one senior editor and 10 correspondents we researched, wrote and published about the various happening of IIT Palakkad from student elections, faculty and alumni interviews, inter IIT fests to new research facilities.
- Proposed an initiative during COVID, to conduct fireside chat with current faculty, ex-faculty and alumni.
- Personally hosted the fireside chats which covered life, academics, profession and philosophy with the rest of the community to foster connections and hope during those uncertain times.

• Student Mentor

July'20 - Dec'20

IIT Palakkad

- Responsible for guiding 10 freshmen focusing on their academic, holistic and overall development in the institute

• Class Representative - Semester II

Jan'19 - May'19

IIT Palakkad

- Responsible for converting student academic grievances and concerns.
- Was elected because of securing Rank 1 in the Mechanical Department in Semester-I